ELECTRICITY DEMAND FORECASTING VIA MULTI-SCALE STRUCTURE-GUIDED TRANSFORMER WITH ADAPTIVE PATHS

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Abstract:

Accurate demand forecasting is crucial for power generation planning, allocation, and management. However, existing methods often overlook power demand uncertainty, long-term dependencies, and feature scaling differences, leading to sub-optimal performance. To overcome these obstacles, this work designs PMS-EDF, a Probabilistic Multi-scale Structure-guided Electricity Demand Forecasting framework with adaptive paths. By integrating a nonlinear neural state space model into a multi-scale Transformer-based structure, PMS-EDF effectively captures uncertainty, temporal dependencies, and feature scaling issues. Evaluations on real-world data show its superior performance and high-quality probabilistic predictions compared to strong baselines.

Keywords:

Electricity demand forecasting; Variant multi-scale Transformer; Adaptive paths

1. Introduction

Extensive research has explored electricity time series forecasting, including demand [1] and price [2]. Despite advancements, instability in energy sources remains. Accurate forecasting is crucial for proactive energy reserves, optimized distribution, and efficient scheduling. This paper focuses on improving electricity demand forecasting (EDF) to address these challenges. Fig. 1 shows the EDF problem.

Currently, primary research methods in the electricity sector include time series models [3], regression models [4], and neural network models [5], [6]. For instance, in [7] the authors proposed a parallel multi-channel network combining convolutional and deep networks for load forecasting in smart grids nowadays. Compared to the stacked CNN-LSTM model, their approach demonstrated higher performance for both day-ahead and hour-ahead forecasts across multiple datasets. GRU and temporal convolutional networks are integrated into short-term power load forecasting in [8], achieving improved performance. A mid-term EDF model is designed in [9] that takes account of economic and seasonal influences.

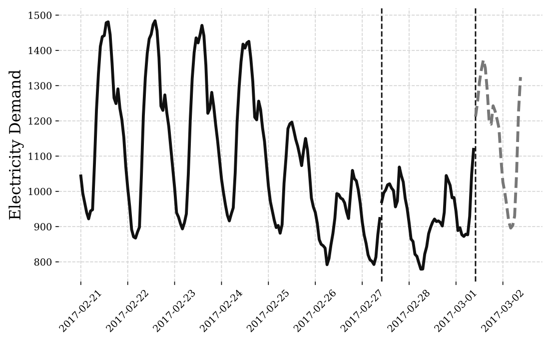

Fig. 1 A visualization of the electricity time series in Panama.

Effective EDF must address the variability of energy sources, especially renewables with randomness and temporal dynamics, and long-term dependencies with external factors at different scales. Traditional architectures often fail to handle these multi-scale challenges.

To address these issues, this work proposes a novel time series forecasting model: Probabilistic Multi-scale Structure-guided Electricity Demand Forecasting (PMS-EDF) with adaptive paths. This model leverages an adaptive multi-scale Transformer structure, incorporating temporal resolution and distance to enhance prediction performance. The model dynamically adjusts to input variations, improving both flexibility and accuracy. A probabilistic state space model further processes data uncertainties, enabling robust probabilistic inference. Empirical evaluations on real-world electricity data show that PMS-EDF achieves superior performance compared to robust baselines in electricity

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demand forecasting.

2. Methodology

Fig. 2 illustrates the overall framework of proposed PMS-EDF model.

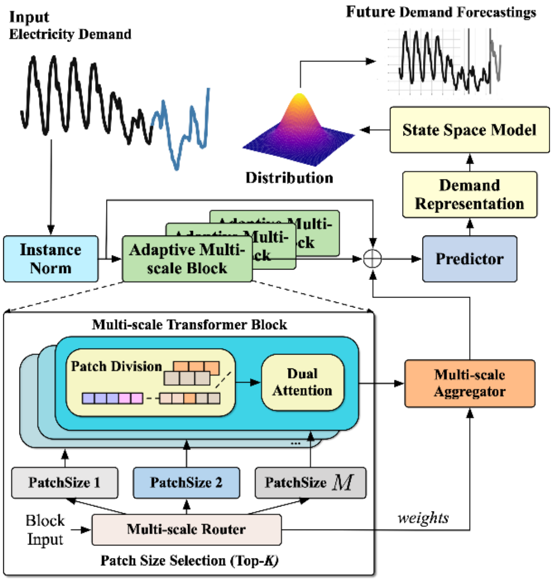

Fig. 2 An overview of PMS-EDF.

2.1. Multi-scale Transformer with Adaptive Pathways

At the core of PMS-EDF is a multi-scale Transformer with adaptive pathways. This module consists of several important components, which are detailed as follows.

Multi-scale Learning Block: The block is powered by Transformer and defines M patch sizes $S = \{s_1, \ldots, s_i, \ldots\}$, where s_i specifies a division of the patches. Let $X \in \mathbb{R}^{\mathrm{L} \times \mathrm{d}}$ be the series of input electricity data with a sequence length of L and an embedding dimension of d. Then we split X into N patches with patch size s_i, $(X^1, X^2, \ldots, X^N)$. Each patch $X^i \in \mathbb{R}^{S \times d}$ contains S time steps. This multi-scale segmentation is paired with a dual attention mechanism to facilitate efficient multi-scale modeling.

Dual Attention: The dual attention mechanism models the temporal dependencies among patches of different scales, comprising both intra-patch and inter-patch attentions. Consider an jth patch $X^j \in \mathbb{R}^{J \times d}$, the patch is transformed as $X^i_{\text{intra}} \in \mathbb{R}^{S \times d'}$, where d' is the size of the embedding. A trainable linear transformation is then applied to X^i_{intra} to generate the keys and values for the attention operation, denoted as K^i_{intra} and V^i_{intra}. The intra query Q^i_{intra} is employed to aggregate the surrounding patches. Local representation is given by:

$$\mathrm{Attn}_{\text{intra}} = \mathrm{Concat}(\mathrm{Attn}^1_{\text{intra}}, \ldots, \mathrm{Attn}^P_{\text{intra}}). \tag{1}$$

The inter-patch attention $\mathrm{Attn}_{\text{inter}}$ is then computed to capture the interactions between patches, which reflect the global correlation in time series:

$$\mathrm{Attn}_{\text{inter}} = \mathrm{SoftMax}\left(Q_{\text{inter}} (K_{\text{inter}})^T / \sqrt{d'_m}\right) \mathrm{V}_{\text{inter}}. \tag{2}$$

Multi-scale Router: The Discrete Fourier Transform (DFT) is implemented to model the input sequence of electricity data into Fourier bases, selecting frequencies that have the highest amplitudes. Subsequently, the patterns X_{sea} is obtained through the inverse DFT, represented as:

$$X_{\text{sea}} = \mathrm{IDFT}\left(\left\{f_1, \ldots, f_{K_f}\right\}, A, \Theta\right), \tag{3}$$

Here, A and Θ represent the amplitude and phase of each frequency obtained from DFT(X). The set $\left\{f_1, \ldots, f_{K_f}\right\}$ corresponds to the frequencies with the highest K_f amplitudes. The complete procedure for calculating the pathway weights $R(X_{\text{trans}})$ is described as follows:

$$\mathrm{SoftMax}\left(X_{\text{trans}} W_r + \epsilon \cdot \mathrm{SoftPlus}(X_{\text{trans}} W_{\text{noise}})\right), \tag{4}$$

where $R(\cdot)$ represents routing function, $\epsilon \sim \mathcal{N}(0,1)$, W_r and $W_{\text{noise}} \in \mathbb{R}^{d \times M}$ are model weights for weighted learning and fusion. To enforce sparsity and focus on critical scales, we keep top-K weights and leave the remaining to 0. The resulting sparse weights are represented as $\bar{R}(X_{\text{trans}})$.

Multi-Scale Aggregator: Given X^i_{out}, which is the multi-scale block we defined above, a transformation function $T_i(\cdot)$ is first applied to align the temporal dimensions across scales. Then the fusion of the multi-scale embeddings is implemented by using the pathways.

$$X_{\text{out}} = \sum_{i=1}^{M} \mathcal{I}\left(\bar{R}(X_{\text{trans}})_i > 0\right) R(X_{\text{trans}})_i \, T_i(X^i_{\text{out}}), \tag{5}$$

here $\mathcal{I}\left(\bar{R}(X_{\text{trans}})_i > 0\right)$ an indicator functions those outputs 1 if $\bar{R}(X_{\text{trans}})_i > 0$, and otherwise 0. This ensures that the patch sizes and block outputs are effectively employed in the multi-scale aggregator.

2.2. Probabilistic State Space Model

The SSM module used in PMS-EDF facilitates the electricity time series learning by providing multiple probabilistic step forecast distributions. It can be expressed as:

$$\begin{cases} L_t = A_t L_{t-1} + B_t + Q_t \epsilon, \\ X_t = C_t L_t + D_t + R_t \epsilon, \end{cases} \tag{6}$$

where $L_t \in \mathbb{R}^{d_s}$ are the latent state at time t, capturing temporal trends and seasonality, ϵ follows the normal distribution, d_s is the state size, A_t are the matrices that update the input embeddings by utilizing historical information. Here variable C_t maps the latent states L_t to observations X_t. Bias terms are represented by B_t and D_t. The vectors in L_t are assumed to be independent, and the positive terms R_t and Q_t are used to express the variances of the observations and latent states. To model nonlinear and conditional probability distributions at each time step, an activation function is used. The above calculation can be formulated as:

$$p(L_0) = \mathcal{N}(L_0|0, I)$$
$$p(L_t) = \mathcal{N}\left(L_t\middle|\delta(A_t L_{t-1} + B_t), \mathrm{diag}(Q_t^2)\right) \quad (7)$$
$$p(X_t) = \mathcal{N}\left(X_t\middle|\delta(C_t L_t + D_t), \mathrm{diag}(R_t^2)\right),$$

where L_0 is the first latent state and $\delta(\cdot)$ represents the Tanh activation function.

3. Experiments

3.1. Datasets and Metrics

The Electricity Demand Forecasting (EDF) performance evaluation of the Panama datasets. Parama datasets record the electricity consumption in Panama for three years starting from 2017, with an interval of one hour. We divide each dataset into train/valid/test sets in a ratio of 2:1:1. The best EDF prediction results are achieved when the validation set loss does not decrease for 50 epochs. Specifically, the electricity demand observation is set to 2 days and the prediction horizon is set to 1 day following [10]. The datasets cover Panama's hourly energy demand and weather data from 2017-2018 and 2019-2020. The power demand ranges from 380.6-1635.9 MW·h (avg: 1194.0) in 2017-2018 and 85.2-1754.9 MW·h (avg: 1217.5) in 2019-2020, accompanied by weather variables including temperature (19.8-27.8°C; 20.22-39.1°C), humidity (17.5-90.6%; 16-100%), and precipitation (0-45.8mm; 0-52.0mm) for the respective periods.

3.2. Baselines

We compare our proposed PMS-EDF with several strong time series models, including statistical models, machine learning approaches, RNN-based approaches, and deep learning approaches. They include HA, ARIMA [11], SVR [12], LSTM and GRU [13], CNN-RNN [14], GRU-VAE [15], DSSM [16], DeepAR [17], PrEF [10], and TPEDF [[18]].

3.3. Results and Discussion

Table 1 shows that PMS-EDF outperforms traditional methods like HA and ARIMA and advanced models such as DeepAR and GRU-VAE by effectively capturing multi-scale temporal dependencies and integrating external features through its adaptive architecture. Unlike DeepAR, which struggles with complex temporal alignments, and GRU-VAE, which has limited uncertainty handling, PMS-EDF leverages its probabilistic design and multi-scale mechanisms to achieve superior accuracy and robustness, as evidenced by an average improvement of 10.72% across metrics. As shown in Fig. 3, the model converges gradually between 50 to 200 epochs. We can see that the variations and differences are relatively small on all three loss curves. This phenomenon indicates that there is no overfitting issue and our model generalizes good to new data.

Table 1 The performance comparison results for electricity demand forecasting

Dataset	Panama 19-20				Panama 17-18			
Metric	MAE(-)	RMSE(-)	CRPS(-)	Log p (+)	MAE (-)	RMSE(-)	CRPS(-)	Log p(+)
HA	86.10	123.7	n/a	n/a	96.79	132.2	n/a	n/a
CNN-RNN	51.74	80.85	0.014	2.784	51.60	72.43	0.012	2.734
ARIMA	80.43	100.6	n/a	n/a	86.05	101.5	n/a	n/a
SVR	58.48	78.30	0.016	2.667	70.77	92.76	0.017	2.453
GRU	51.84	74.34	0.016	2.680	57.49	80.24	0.015	2.677
DSSM	51.87	71.21	0.013	2.793	43.66	64.14	0.010	2.801
LSTM	52.04	73.60	0.015	2.694	56.68	79.24	0.015	2.649
DeepAR	46.03	67.87	0.015	2.753	45.49	66.85	0.011	2.764
GRU-VAE	52.18	73.74	0.015	2.675	60.24	76.04	0.013	2.727
PrEF	42.01	63.94	0.012	**2.851**	40.84	66.00	0.008	3.118
TPEDF	41.40	56.03	0.007	2.773	39.56	51.10	0.007	3.170
OURS	**37.60**	**48.13**	**0.006**	2.830	**35.54**	**45.99**	**0.005**	**3.177**
Improv.	-3.8	-7.9	-0.001	-0.021	-4.02	-5.11	-0.002	+0.007

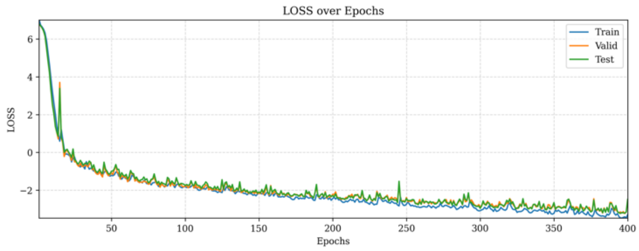

Fig. 3 Loss change curves.

4. Conclusions

This work proposes a probabilistic electricity demand forecasting framework using a multi-scale Transformer with adaptive paths. By integrating a nonlinear state space model, it captures uncertainty, temporal dependencies, and feature scaling issues. Experiments on real-world datasets demonstrate high predictive accuracy. Future work aims to improve modeling of randomness and interference while addressing the framework's high computational costs to enhance scalability and efficiency for large-scale systems.

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